NCLT Dataset

The University of Michigan North Campus Long-Term Vision and LIDAR Dataset.

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Overview

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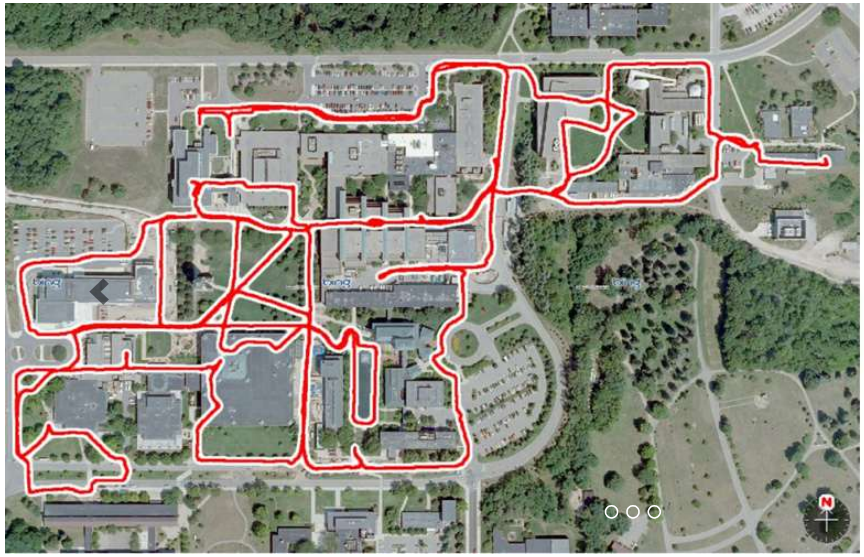

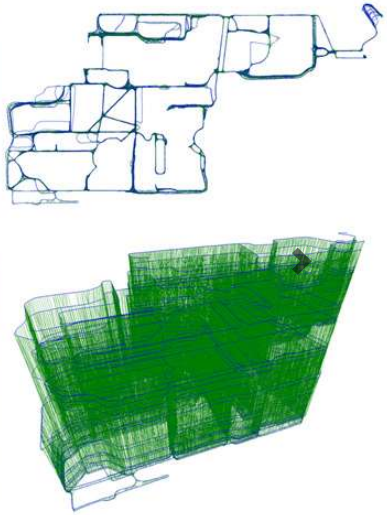

Sensors
Velodyne HDL-32E LIDAR
Ladybug3 Omnidirectional Vision
Microstrain GX3 IMU
KVH Fiber Optic Gyro
Wheel Odometry
Standard and RTK GPS

Data Collection
27 Sessions: Varying route and time of day
15 Months: Jan. 2012 to April 2013
Session Duration: ≈ 1.29 hours
Session Distance: ≈ 5.5 km
Total Duration: 34.9 hours
Total Distance: 147.4 km

Documentation

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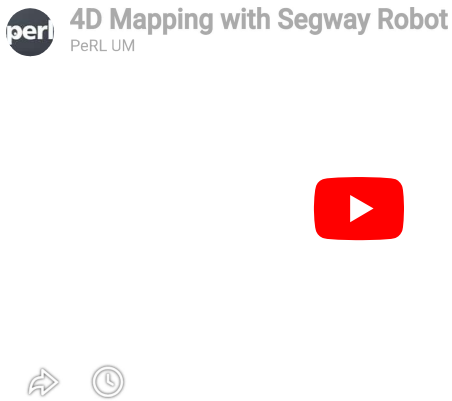
Download the PDF (nclt.pdf (nclt.pdf)), which provides a detailed description of the dataset. If you use this dataset in your research please cite:

Nicholas Carlevaris-Bianco, Arash K. Ushani, and Ryan M. Eustice, *University of Michigan North Campus Long-Term Vision and Lidar Dataset*, International Journal of Robotics Research, 2016. [BibTeX (http://robots.engin.umich.edu/Publications?action=bibentry&bibfile=perl.bib&bibref=ncarlevaris-2015a#ncarlevaris-2015aBib)]

(nclt.pdf)

nclt.pdf (nclt.pdf)



Changelog

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Feburary 4, 2017: The covariances for keyframes in the ground truth pose estimate has been added. These marginal covariances are extracted from the SLAM graph and recorded in the same format as other covariances in the dataset.

January 16, 2017: In Table 6, the x and y columns for the camera center coordinates is swapped. The correct values are given in cam_params.zip (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/ladybug3_calib/cam_params.zip). Additionally, a python script demonstrating how to use the camera parameter files and project velodyne points to images has been added.

August 14, 2018: Sensor data from the KVH fiber-optic gyro and the left and right wheel velocities have been added to the sen.tar.gz files.

March 24, 2019: The ground truth pose was generated using a SLAM graph, and then interpolation using odometry was used to provide this at roughly 100 Hz. A script showing how to inspect only the ground truth pose from the nodes in the SLAM graph has been added. Thanks to Alexander Schaefer for helping us discover the ambiguity regarding this subtlety.

License

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The NCLT Dataset is made available under the Open Database License (available here) (http://opendatacommons.org/licenses/odbl/1.0/). Any rights in individual contents of the database are licensed under the Database Contents License (available here) (http://opendatacommons.org/licenses/dbcl/1.0/).

In short, this means that you are free to use this dataset, share, create derivative works, or adapt it, as long as you credit our work, offer any publically used adapted version of this dataset under the same license, and keep any redistribution of this dataset open.

When using the NCLT Dataset, please credit our work by citing the paper above.

Download

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Data

Session	Images	Sensors	Velodyne	H
2012-01-08	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-01-08_lb3.tar.gz) (106 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-01-08_sen.tar.gz) (114 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-01-08_vel.tar.gz) (16 GB)	h 2. 01
2012-01-15	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-01-15_lb3.tar.gz) (130 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-01-15_sen.tar.gz) (136 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-01-15_vel.tar.gz) (20 GB)	h 2. 01
2012-01-22	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-01-22_lb3.tar.gz) (92 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-01-22_sen.tar.gz) (107 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-01-22_vel.tar.gz) (15 GB)	h 2. 01
2012-02-02	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-02-02_lb3.tar.gz) (105 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-02-02_sen.tar.gz) (119 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-02-02_vel.tar.gz) (16 GB)	h 2. 02
2012-02-04	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-02-04_lb3.tar.gz) (85 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-02-04_sen.tar.gz) (96 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-02-04_vel.tar.gz) (13 GB)	h 2. 02
2012-02-05	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-02-05_lb3.tar.gz) (109 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-02-05_sen.tar.gz) (114 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-02-05_vel.tar.gz) (16 GB)	h 2. 02
2012-02-12	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-02-12_lb3.tar.gz) (101 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-02-12_sen.tar.gz) (105 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-02-12_vel.tar.gz) (15 GB)	h 2. 02
2012-02-18	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-02-18_lb3.tar.gz) (102 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-02-18_sen.tar.gz) (108 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-02-18_vel.tar.gz) (15 GB)	h 2. 02
2012-02-19	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-02-19_lb3.tar.gz) (101 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-02-19_sen.tar.gz) (109 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-02-19_vel.tar.gz) (15 GB)	h 2. 02

Session	Images	Sensors	Velodyne	H
2012-03-17	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-03-17_lb3.tar.gz) (100 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-03-17_sen.tar.gz) (101 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-03-17_vel.tar.gz) (13 GB)	h 2. 03
2012-03-25	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-03-25_lb3.tar.gz) (107 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-03-25_sen.tar.gz) (98 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-03-25_vel.tar.gz) (13 GB)	h 2. 03
2012-03-31	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-03-31_lb3.tar.gz) (95 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-03-31_sen.tar.gz) (107 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-03-31_vel.tar.gz) (95 GB)	h 2. 03
2012-04-29	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-04-29_lb3.tar.gz) (87 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-04-29_sen.tar.gz) (53 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-04-29_vel.tar.gz) (8 GB)	h 2. 04
2012-05-11	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-05-11_lb3.tar.gz) (105 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-05-11_sen.tar.gz) (104 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-05-11_vel.tar.gz) (14 GB)	h 2. 05
2012-05-26	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-05-26_lb3.tar.gz) (87 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-05-26_sen.tar.gz) (108 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-05-26_vel.tar.gz) (15 GB)	h 2. 05
2012-06-15	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-06-15_lb3.tar.gz) (94 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-06-15_sen.tar.gz) (67 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-06-15_vel.tar.gz) (10 GB)	h 2. 06
2012-08-04	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-08-04_lb3.tar.gz) (90 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-08-04_sen.tar.gz) (97 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-08-04_vel.tar.gz) (14 GB)	h 2. 08
2012-08-20	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-08-20_lb3.tar.gz) (89 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-08-20_sen.tar.gz) (101 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-08-20_vel.tar.gz) (15 GB)	h 2. 08
2012-09-28	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-09-28_lb3.tar.gz) (84 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-09-28_sen.tar.gz) (94 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-09-28_vel.tar.gz) (13 GB)	h 2. 09
2012-10-28	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-10-28_lb3.tar.gz) (98 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-10-28_sen.tar.gz) (104 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-10-28_vel.tar.gz) (14 GB)	h 2. 10
2012-11-04	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-11-04_lb3.tar.gz) (88 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-11-04_sen.tar.gz) (97 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-11-04_vel.tar.gz) (13 GB)	h 2. 11
2012-11-16	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-11-16_lb3.tar.gz) (89 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-11-16_sen.tar.gz) (98 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-11-16_vel.tar.gz) (13 GB)	h 2. 11
2012-11-17	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-11-17_lb3.tar.gz) (107 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-11-17_sen.tar.gz) (107 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-11-17_vel.tar.gz) (14 GB)	h 2. 11

Session	Images	Sensors	Velodyne	H
2012-12-01	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2012-12-01_lb3.tar.gz) (94 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2012-12-01_sen.tar.gz) (93 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2012-12-01_vel.tar.gz) (12 GB)	h 2. 12
2013-01-10	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2013-01-10_lb3.tar.gz) (37 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2013-01-10_sen.tar.gz) (21 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2013-01-10_vel.tar.gz) (3 GB)	h 2. 01
2013-02-23	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2013-02-23_lb3.tar.gz) (88 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2013-02-23_sen.tar.gz) (97 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2013-02-23_vel.tar.gz) (15 GB)	h 2. 02
2013-04-05	lb3.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/images/2013-04-05_lb3.tar.gz) (88 GB)	sen.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/sensor_data/2013-04-05_sen.tar.gz) (84 MB)	vel.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/velodyne_data/2013-04-05_vel.tar.gz) (13 GB)	h 2. 04

Python Downloader

For systems with python and `wget`, we provide a downloader script (downloader.py (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/downloader.py)) that can be used to download different subsets of the data. In this script `wget` calls are made with the `--continue` flag, which allows you to restart a download if the connection is interrupted. Example usage:

Download the entire dataset

```
./downloader.py --all
```

Download all data from 2012-01-08

```
./downloader.py --date="2012-01-08" --all
```

Download all sensor data

```
./downloader.py --sen
```

Download imagery from 2012-01-08

```
./downloader.py --date="2012-01-08" --lb3
```

Download imagery and Velodyne data from 2012-01-08

```
./downloader.py --date="2012-01-08" --lb3 --vel
```

LadyBug3 Parameters and Undistort Maps

Camera Parameters	cam_params.zip (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/ladybug3_calib/cam_params.zip) (2 KB)	
All Maps	U2D_ALL_1616X1232.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/ladybug3_calib/U2D_ALL_1616X1232.tar.gz) (151 MB)	D2U_ALL_1616X1232.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/ladybug3_calib/D2U_ALL_1616X1232.tar.gz) (148 MB)
Cam 0	U2D_Cam0_1616X1232.txt (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/ladybug3_calib/U2D_Cam0_1616X1232.txt) (61 MB)	D2U_Cam0_1616X1232.txt (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/ladybug3_calib/D2U_Cam0_1616X1232.txt) (61 MB)

Cam 1	U2D_Cam1_1616X1232.txt (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/ladybug3_calib/U2D_Cam1_1616X1232.txt) (61 MB)	D2U_Cam1_1616X1232.txt (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/ladybug3_calib/D2U_Cam1_1616X1232.txt) (61 MB)
Cam 2	U2D_Cam2_1616X1232.txt (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/ladybug3_calib/U2D_Cam2_1616X1232.txt) (61 MB)	D2U_Cam2_1616X1232.txt (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/ladybug3_calib/D2U_Cam2_1616X1232.txt) (61 MB)
Cam 3	U2D_Cam3_1616X1232.txt (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/ladybug3_calib/U2D_Cam3_1616X1232.txt) (61 MB)	D2U_Cam3_1616X1232.txt (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/ladybug3_calib/D2U_Cam3_1616X1232.txt) (61 MB)
Cam 4	U2D_Cam4_1616X1232.txt (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/ladybug3_calib/U2D_Cam4_1616X1232.txt) (61 MB)	D2U_Cam4_1616X1232.txt (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/ladybug3_calib/D2U_Cam4_1616X1232.txt) (61 MB)
Cam 5	U2D_Cam5_1616X1232.txt (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/ladybug3_calib/U2D_Cam5_1616X1232.txt) (61 MB)	D2U_Cam5_1616X1232.txt (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/ladybug3_calib/D2U_Cam5_1616X1232.txt) (61 MB)

Sample Python Scripts

File	Description
python.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/python/python.tar.gz)	All sample python scripts
read_gps.py (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/python/read_gps.py)	Read GPS from data file
read_ground_truth.py (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/python/read_ground_truth.py)	Read ground truth from data file
read_hokuyo_30m.py (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/python/read_hokuyo_30m.py)	Read Hokuyo UTM-30LX from data file
read_hokuyo_4m.py (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/python/read_hokuyo_4m.py)	Read Hokuyo URG-04LX from data file
read_ms25.py (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/python/read_ms25.py)	Read Microstrain data from data file
read_ms25_euler.py (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/python/read_ms25_euler.py)	Read Microstrain raw Euler angles from data file
read_odom.py (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/python/read_odom.py)	Read odometry from data file
read_vel_hits.py (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/python/read_vel_hits.py)	Read Velodyne data from data file
read_vel_sync.py (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/python/read_vel_sync.py)	Read one revolution of Velodyne from data file
project_vel_to_cam.py (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/python/project_vel_to_cam.py)	Project velodyne points to camera image
Undistort.py (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/python/undistort.py)	Undistort LadyBug3 images
gps_to_rosbag.py (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/python/gps_to_rosbag.py)	Convert GPS file to ROSbag
hokuyo_30m_to_rosbag.py (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/python/hokuyo_30m_to_rosbag.py)	Convert Hokuyo UTM-30LX file to ROSbag

File	Description
hokuyo_4m_to_rosbag.py (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/python/hokuyo_4m_to_rosbag.py)	Convert Hokuyo URG-04LX file to ROSbag
ms25_euler_to_rosbag.py (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/python/ms25_euler_to_rosbag.py)	Convert Microstrain Euler file to ROSbag
ms25_to_rosbag.py (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/python/ms25_to_rosbag.py)	Convert Microstrain file to ROSbag
vel_to_rosbag.py (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/python/vel_to_rosbag.py)	Convert Velodyne file to ROSbag
sensordata_to_rosbag.py (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/python/sensordata_to_rosbag.py)	Convert all sensor data to ROSbag

Other Files

File	Description
segway_cad.tar.gz (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/segway_cad.tar.gz)	Solidworks CAD Model of the Segway
laser_angles.csv (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/laser_angles.csv)	Laser angles for the Velodyne
DSP3000_Manual.pdf (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/manuals/DSP3000_Manual.pdf)	KVH DSP-3000 Fiber Optic Gyro Technical Manual
URG-04LX_spec_en.pdf (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/manuals/URG-04LX_spec_en.pdf)	Scanning Laser Range Finder URG-04LX Specification
UTM-30LX_spec_en.pdf (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/manuals/UTM-30LX_spec_en.pdf)	Scanning Laser Range Finder UTM-30LX/LN Specification
GPS_18x_Tech_Specs.pdf (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/manuals/GPS_18x_Tech_Specs.pdf)	GPS 18x Technical Specifications
3DM-GX3-45-Theory_of_Operation.pdf (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/manuals/3DM-GX3-45_Theory_of_Operation.pdf)	3DM-GX3-45 Theory of Operation
Ladybug3-Technical-Reference.pdf (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/manuals/Ladybug3-Technical-Reference.pdf)	Ladybug3 Technical Reference Manual
HDL-32E_manual.pdf (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/manuals/HDL-32E_manual.pdf)	User's Manual and Programming Guide: HDL-32E High Definition LiDAR Sensor
dl4_manual.pdf (https://s3.us-east-2.amazonaws.com/nclt.perl.engin.umich.edu/manuals/dl4_manual.pdf)	DL-4plus User Manual